

***Generation Interconnection
Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AD2-001***

Grangston 138 kV

December 2019

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

General

Big Sandy Peaker Plant, LLC proposes to increase generation the existing Big Sandy Peaker Gas Turbine Plant connected to the Grangston 138 kV Substation by 19.0 MW (19.0 MW Capacity) in Wayne County, West Virginia (see Figure 2). Total output of the plant will be 319.0 MW (319.0 MW Capacity). The AD2-001 uprate request is due to additional machine capability and does not reflect any electrical or mechanical changes to the plant.

It was determined that this project does not qualify for an accelerated study since there are other projects in the area and Stability Analysis and a Reactive Power assessment will be required. The Stability Analysis and a Reactive Power assessment are not performed until the System Impact Study Phase.

The requested in service date is February 1, 2018.

Point of Interconnection

Point of Interconnection (Grangston 138kV substation)

Attachment facilities not required for an existing facility.

Note: No changes to the existing 138 kV revenue metering is required for the additional generation.

Cost Summary

The AD2-001 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Allocation for New System Upgrades	\$ 0
Contribution for Previously Identified Upgrades	\$ 0
Total Costs	\$ 0

Attachment Facilities

There are no Attachment Facilities required to be constructed by the Transmission Owner.

Direct Connection Cost Estimate

There are no Direct Connection Facilities required to be constructed by the Transmission Owner.

Non-Direct Connection Cost Estimate

There are no Non-Direct Connection Facilities required to be constructed by the Transmission Owner.

Schedule

It is anticipated that the time between receipt of executed agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would be between 24 to 36 months after signing an interconnection agreement.

Note: The time provided between anticipated normal completion of System Impact, Facilities Studies, subsequent execution of ISA and ICSA documents, and the proposed Backfeed Date is shorter than usual and may be difficult to achieve.

Interconnection Customer Requirements

It is understood that Big Sandy Peaker Plant, LLC is responsible for all costs associated with this interconnection. The costs above are reimbursable to AEP. The cost of Big Sandy Peaker Plant, LLC's generating plant and the costs for the line connecting the generating plant to the Grangston 138 kV substation are not included in this report; these are assumed to be Big Sandy Peaker Plant, LLC's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for American Electric Power to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

AEP Requirements

The Interconnection Customer will be required to comply with all AEP Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "Requirements for Connection of New Facilities or Changes to Existing Facilities Connected to the AEP Transmission System" document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-aep/aep-interconnection-requirements.ashx>

Network Impacts

The Queue Project AD2-001 was evaluated as a 19.0 MW (Capacity 19.0 MW) injection into the Grangston 138 kV substation in the AEP area. Project AD2-001 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD2-001 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None.

Light Load Analysis - 2021

Light Load Studies to be conducted during the System Impact Study phase (as required by PJM Manual 14B).

Multiple Facility Contingency

(Double Circuit Tower Line contingencies were studied for the full energy output. The contingencies of Line with Failed Breaker and Bus Fault will be performed for the Impact Study.)

None.

Short Circuit

(Summary of impacted circuit breakers)

None.

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None.

Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

None.

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

See Attachment 3

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None.

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None.

Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

None.

Incremental Capacity Transfer Rights (ICTRs)

Will be determined at a later study phase

Affected System Analysis & Mitigation

LGEE Impacts:

LGEE Impacts to be determined during later study phases (as applicable).

MISO Impacts:

MISO Impacts to be determined during later study phases (as applicable).

Duke, Progress & TVA Impacts:

Duke Carolina, Progress, & TVA Impacts to be determined during later study phases (as applicable).

OVEC Impacts:

OVEC Impacts to be determined during later study phases (as applicable).

Figure 1: Point of Interconnection (Grangston 138kV Substation)
Single-Line Diagram

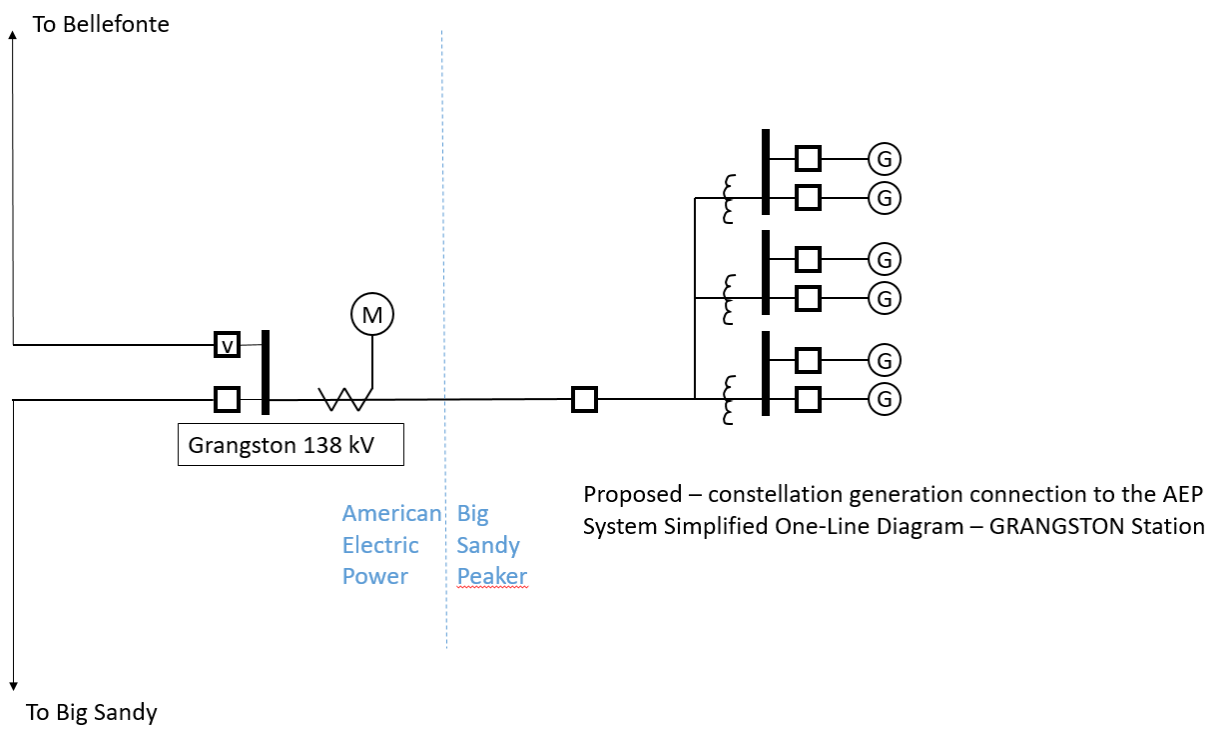


Figure 2: Point of Interconnection (Grangston 138kV Substation)



Figure 3: Dynamic Simulation Analysis

Executive Summary

Generator Interconnection Request AD2-001 is for a 19.0 MW uprate of the existing Big Sandy Peaker Gas Turbine Plant. The uprate increases the Maximum Facility Output (MFO) of the facility from 300.0 MW to 319.0 MW due to additional machine capability and does not reflect any electrical or mechanical changes to the plant. The details of the uprate is given in Table 1. AD2-001 consists of 6 gas turbine 52.2 MW to 54.00 MW generators with a Point of Interconnection (POI) at the Grangston 138 kV (AD2-001 POI) substation, in the American Electric Power (AEP) transmission system, Wayne County, West Virginia.

This report describes a dynamic simulation analysis of AD2-001 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2021 light load case, modified to include applicable queue projects. AD2-001 has been dispatched online at maximum power output, with 0.95 p.u. voltage at the generator bus.

AD2-001 was tested for compliance with NERC, PJM, Transmission Owner and other applicable criteria. Steady-state condition and 47 contingencies were studied, each with a 20 second simulation time period. Studied faults included:

- a) Steady state operation (20 second);
- b) Three-phase faults with normal clearing time and high speed reclosing (HSR);
- c) Single-phase faults with single phase stuck breaker;
- d) Single-phase faults placed at 80% of the line with delayed (Zone 2) clearing at line end remote from the fault due to primary communications/relay failure;

No relevant bus or tower faults were identified for this study.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For all of the fault contingencies tested on the 2021 light load case:

- a) AD2-001 was able to ride through the faults except for faults where protective action trips the generator.
- b) AD2-001 Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- c) Following fault clearing, all bus voltages recovered to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element tripped, other than those either directly connected or designed to trip as a consequence of that fault.

AD2-001 units were found to NOT meet their reactive power requirements.